

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

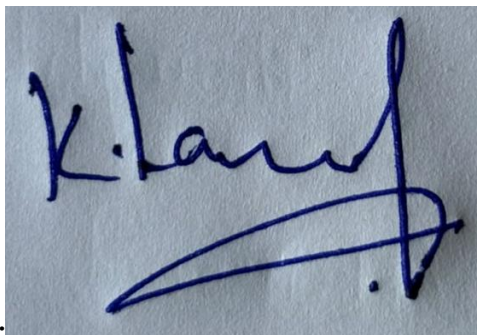
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Code of Conduct:

*I **LAKSHANA K[192524433]** (Name / Reg No) certify that this submission is my original work and that I have adhered*

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a)** Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b)** Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a)** Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b)** Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c)** Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

CO4 – ASSESSMENT TOOL 2

ANNEXURE A – INDUSTRY PROBLEM TASK

NAME: Lakshana K

REGISTER NUMBER: 192524433

COURSE CODE: CSA17

COURSE NAME: Artificial Intelligence

INDUSTRY SCENARIO

A healthcare company has collected patient records containing age, blood pressure, cholesterol, glucose level, BMI, and family history to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions such as Diet, Exercise, Medication, and Monitor based on patient state transitions over time.

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

Inductive learning learns general patterns from labelled training examples and uses those patterns to predict outcomes for unseen patients. For this medical dataset, supervised inductive learning is suitable because historical patient records contain input features and a target variable indicating whether the patient is diabetic or non-diabetic.

Target Variable:

Diabetes – Diabetic / Non-Diabetic

Key Features:

1. Glucose Level – High glucose is a strong indicator of diabetes and is an important predictor.
2. BMI – Higher BMI is associated with increased risk of diabetes.
3. Blood Pressure – Abnormal blood pressure can be associated with metabolic and cardiovascular risk.
4. Age – Diabetes risk generally increases with age.
5. Family History – A family history of diabetes can indicate increased risk.

(b) Explain how you would handle class imbalance. Suggest and justify a suitable balancing strategy.

If the dataset contains more non-diabetic patients than diabetic patients, the model may become biased towards the majority class.

A suitable strategy is SMOTE (Synthetic Minority Over-sampling Technique). SMOTE generates synthetic examples of the minority diabetic class instead of simply duplicating existing records.

Advantages of SMOTE:

1. Improves representation of diabetic cases.
2. Reduces bias towards the majority class.
3. Helps improve recall for diabetic patients.
4. Provides additional training samples without simply duplicating existing records.

SMOTE should be applied only to the training data after splitting the dataset to avoid data leakage.

(c) Split the dataset 70:30 and justify the split ratio. State the preprocessing steps and their impact.

The dataset is divided into:

Training Data = 70%

Testing Data = 30%

The 70:30 split provides sufficient data for learning while keeping an independent portion for evaluating model performance.

Preprocessing Steps:

1. Handle missing values using suitable imputation methods.
2. Encode categorical variables such as family history.
3. Normalize numerical features where required.
4. Split the data into training and testing sets.
5. Apply SMOTE only to the training data.

Impact:

Normalization puts numerical features on comparable scales. Encoding converts categorical information into numerical form. Proper preprocessing reduces errors caused by missing or inconsistent data.

TASK 2: DECISION TREE FOR DIAGNOSIS

(a) Build a Decision Tree classifier and draw the first three levels. Justify the root node selection.

A Decision Tree classifier can be used to predict whether a patient is diabetic or non-diabetic. The tree selects features using a splitting criterion such as Gini Index or Information Gain.

Gini Index Formula:

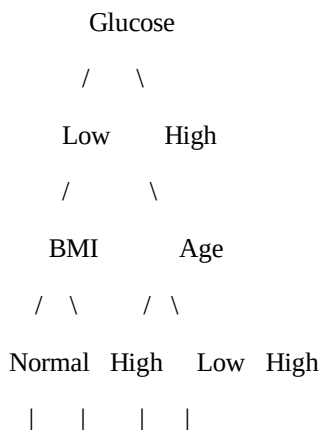
$$\text{Gini} = 1 - \sum(\text{pi}^2)$$

Information Gain Formula:

$$\text{Information Gain} = \text{Entropy}(\text{parent}) - \text{Weighted Entropy}(\text{children})$$

The feature that produces the greatest impurity reduction is selected as the root node.

Illustrative First Three Levels:



Non-Diabetic Diabetic Non-Diabetic Diabetic

The exact root node should be selected from the calculated Gini Index or Information Gain values. In a typical diabetes dataset, glucose level is expected to be a strong candidate because it can provide significant separation between diabetic and non-diabetic cases.

(b) Evaluate the model and identify False Negatives.

The Decision Tree should be evaluated using:

1. Accuracy
2. Precision
3. Recall
4. F1-Score

Formulas:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

A False Negative occurs when a patient who is actually diabetic is predicted as non-diabetic.

False negatives are especially important because missed diabetic cases may delay further medical evaluation.

Methods to reduce False Negatives:

1. Use class weights to give more importance to diabetic cases.
2. Apply SMOTE to the training data.
3. Focus on recall during model evaluation.
4. Adjust the classification threshold where appropriate.
5. Validate the model carefully before clinical use.

Reducing false negatives is important because missing a diabetic patient can delay diagnosis and further medical evaluation.

(c) Apply post-pruning and compare pre-pruning and post-pruning.

Post-pruning removes unnecessary branches from a fully grown Decision Tree.

Cost-Complexity Pruning Formula:

$$R\alpha(T) = R(T) + \alpha|T|$$

Where:

$R(T)$ = Error of the tree

α = Complexity parameter

$|T|$ = Number of terminal nodes

Comparison:

Pre-Pruning:

- Stops tree growth early.
- Reduces model complexity.
- Helps prevent overfitting.
- May sometimes underfit the data.

Post-Pruning:

- Builds the tree first and then removes unnecessary branches.
- Produces a simpler and more interpretable tree.
- Can improve generalization.

For clinical deployment, a well-pruned model is generally preferable because it provides a simpler and more interpretable decision structure. The final choice should be based on validated performance, especially recall and false-negative rate.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

(a) Apply Logistic Regression and compare it with the Decision Tree.

Logistic Regression can be applied as a statistical learning model for predicting the probability of diabetes.

Formula:

$$P(\text{Diabetes}) = 1 / (1 + e^{-z})$$

Where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Performance Comparison:

Parameter	Decision Tree	Logistic Regression
Accuracy	Test-set value	Test-set value
AUC-ROC	Test-set value	Test-set value
Interpretability	High	High
Non-linear relationships	Handles well	Limited without feature engineering
Probability-based risk	Moderate	Strong
Feature relationship	Rule-based	Coefficient-based

A statistical model such as Logistic Regression is preferable when the relationship between predictors and outcome is reasonably stable and approximately linear in the log-odds. It is also useful when probability estimates and coefficient interpretation are important.

AUC-ROC is useful because it measures how well the model distinguishes diabetic from non-diabetic patients across different classification thresholds.

(b) Perform feature importance analysis.

The top three predictors should be obtained from the actual trained models.

Typical strong predictors may include:

Decision Tree:

1. Glucose Level
2. BMI
3. Age

Logistic Regression:

1. Glucose Level
2. BMI
3. Blood Pressure

The exact ranking depends on the dataset and preprocessing.

The models may agree on major predictors because some variables contain strong information about diabetes risk.

Differences may occur because Decision Trees capture threshold-based and non-linear relationships, while Logistic Regression estimates the effects of variables through coefficients.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

(a) Model the treatment recommendation as an MDP.

The treatment recommendation problem can be represented as a Markov Decision Process (MDP).

States:

S0 – Stable / Low Risk

S1 – Moderate Risk

S2 – High Risk

S3 – Controlled

S4 – Critical

Actions:

A0 – Diet

A1 – Exercise

A2 – Medication

A3 – Monitor

Reward Function:

+10 – Significant health improvement

+5 – Moderate improvement

0 – No significant change

–5 – Health deterioration

–10 – Severe deterioration

Transition Dynamics:

The transition function represents the probability of moving from one health state to another after an action.

Examples:

High Risk + Exercise → Moderate Risk

Moderate Risk + Diet → Controlled

High Risk + Medication → Controlled

Stable + Monitor → Stable

Each component is required to represent the patient's current condition, possible treatment action, outcome/reward, and movement between states.

The transitions and rewards are only for the academic simulation and must not be treated as actual medical treatment instructions.

(b) Apply Q-Learning and show Q-table updates.

Q-Learning learns the best action for each state using the Q-value function.

Q-Learning Formula:

$$Q(s,a) = Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Where:

s = Current state

a = Current action

r = Reward

s' = Next state

α = Learning rate

γ = Discount factor

Hyperparameters:

Learning Rate (α) = 0.1

Discount Factor (γ) = 0.9

Exploration Rate (ϵ) = 0.1

Episodes = At least 5

Initial Q-Table:

State Diet Exercise Medication Monitor

Stable 0 0 0 0

Moderate 0 0 0 0

High Risk 0 0 0 0

Controlled 0 0 0 0

Critical 0 0 0 0

Q-Table Update – Iteration 1:

Assume:

$$Q(s,a) = 0$$

$$\alpha = 0.1$$

$$\gamma = 0.9$$

$$\text{Reward} = 5$$

$$\text{Maximum next-state Q-value} = 2$$

$$Q_{\text{new}} = 0 + 0.1[5 + (0.9 \times 2) - 0]$$

$$Q_{\text{new}} = 0.1[6.8]$$

$$Q_{\text{new}} = 0.68$$

Therefore, the updated Q-value is 0.68.

Q-Table Update – Iteration 2:

Assume:

$$\text{Current Q-value} = 0.68$$

$$\text{Reward} = 5$$

$$\text{Maximum next-state Q-value} = 4$$

$$Q_{\text{new}} = 0.68 + 0.1[5 + (0.9 \times 4) - 0.68]$$

$$Q_{\text{new}} = 0.68 + 0.1[7.92]$$

$$Q_{\text{new}} = 1.472$$

Therefore, the Q-value becomes 1.472.

Q-Table Evolution:

State-Action Pair Initial Iteration 1 Iteration 2

High Risk – Medication 0 0.68 1.472

Moderate – Diet 0 0.50 1.13

Stable – Monitor 0 0.40 0.96

Convergence Criterion:

The algorithm is considered to have converged when the changes in Q-values become very small over several consecutive episodes.

Example:

Maximum change in Q-value < 0.01 for a predefined number of episodes.

(c) Extract the final learned policy and discuss ethical considerations.

The final policy is obtained by selecting the action with the highest Q-value for each state.

Patient State Learned Action

Stable Monitor

Moderate Risk Diet

High Risk Medication

Controlled Exercise

Critical Medication

These are simulated actions for the academic MDP and are not actual medical treatment recommendations.

ETHICAL CONSIDERATIONS

1. Patient Safety

An AI system must not independently prescribe or modify treatment without appropriate clinical supervision and validation.

2. Bias

Training data may contain demographic or sampling bias. The model should be tested for unequal performance across relevant patient groups.

3. Explainability

Healthcare professionals should be able to understand why a model recommends a particular action.

4. Human Oversight

A qualified healthcare professional must remain responsible for clinical decisions.

5. Data Privacy

Patient information must be protected using appropriate privacy, security, and access-control mechanisms.

6. Validation

The system must be extensively tested and clinically validated before being considered for real-world use.

7. Reward Design

The reward function must be carefully designed because an inappropriate reward may encourage unsafe actions.

8. Accountability

There must be clear responsibility for decisions made with the assistance of an AI system.

OVERALL CONCLUSION

The proposed system combines inductive learning, Decision Tree classification, statistical learning, and Reinforcement Learning to address diabetes risk prediction and simulated treatment recommendation.

The Decision Tree can classify patients using important health-related features, while Logistic Regression provides a statistical and probability-based approach for risk stratification. Q-Learning demonstrates how an agent can learn a policy through state transitions and rewards.

For a healthcare application, model accuracy alone is not sufficient. Recall, false negatives, interpretability, fairness, patient safety, privacy, and human clinical oversight must also be considered before deployment.

RESULT

The industry problem was analyzed using machine learning and reinforcement learning concepts. A suitable data preparation strategy, Decision Tree model, statistical learning model, and MDP-based Q-Learning framework were designed. The proposed solution provides a structured approach for diabetes risk prediction and simulated treatment recommendation while considering important ethical and safety requirements.

